

Geographic Patterns in County-Level PM2.5 Monitoring Data Across Wisconsin, 2024

Research question

The core question is where monitored county-level PM2.5 means were highest in Wisconsin during 2024 and whether those monitored counties showed meaningful spatial clustering.

Data and methods

The rebuild combines a 72-county TIGER/Line boundary layer, user-supplied Wisconsin PM2.5 summary files, and benchmark county means preserved in the rebuild prompt. County FIPS codes were standardized, unmonitored counties were retained on statewide maps, and spatial diagnostics were run under both the legacy Queen specification and K-nearest-neighbor sensitivity specifications with k values of 3, 4, and 5.

Main findings

Sixteen monitored counties appear in the legacy benchmark snapshot. Grant County is the highest monitored county and Monroe County the lowest. The descriptive correlation between PM2.5 and log population density is 0.604, but that relationship should be read cautiously given the small monitored sample and the non-random placement of monitors. Across Queen, KNN3, KNN4, and KNN5, Global Moran's I remains statistically insignificant.

Spatial-statistics caveat

Spatial statistics depend on how geographic relationships are encoded. Under Queen contiguity, the monitored-county network separated into eight components and left five counties without monitored contiguous neighbors. I therefore evaluated K-nearest-neighbor specifications with k values of 3, 4, and 5. Although the magnitude of Global Moran's I varied across specifications, none of the permutation tests was statistically significant. Local LISA labels were less stable: classifications appearing under one or two specifications generally disappeared when the neighborhood definition changed. This sensitivity suggests that county-specific clusters should be treated as exploratory, especially given the limited monitoring coverage.

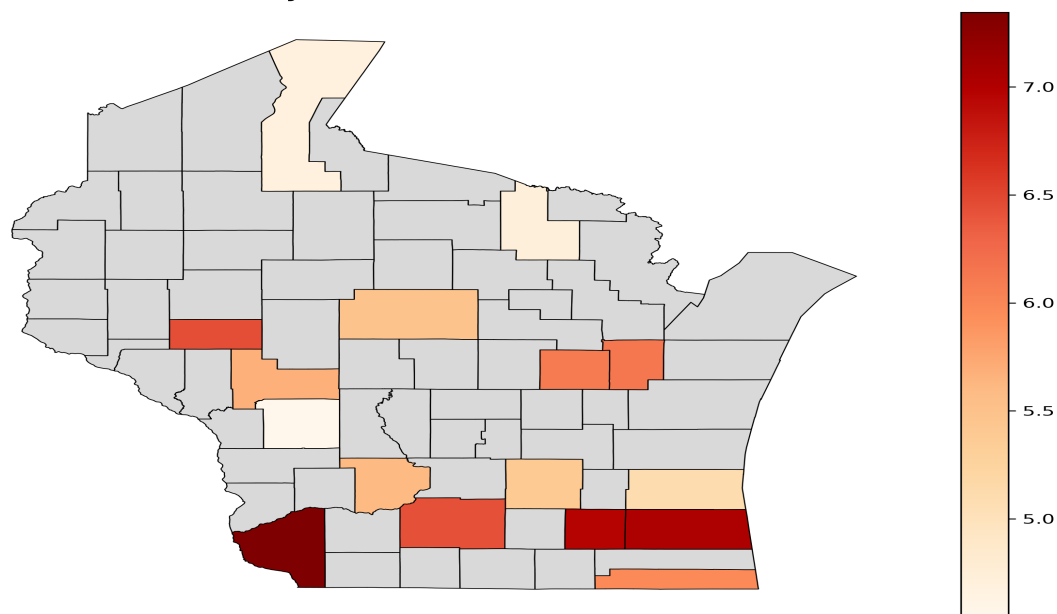
Limitations

This remains a monitor-based county snapshot rather than a statewide exposure surface. KNN does not repair sparse coverage, it only changes how the monitored counties are linked in the robustness check. Missing raw daily observations also prevent full record-level replication of the original classroom workflow.

Policy relevance and future work

The project is most useful as a transparent demonstration of reproducible geospatial workflow, careful source documentation, and cautious interpretation of spatial-statistical sensitivity rather than as evidence of confirmed county clusters.

Wisconsin County-Level Annual Mean PM2.5 (2024)



None of the global permutation tests is statistically significant at $\alpha = 0.05$.

Global Moran's I Across Spatial-Weight Specifications

